

Problem

Rework Example 16.9 for the circuit shown in Fig. 16.20.

Example 16.9:

For the s -domain circuit in Fig. 16.19, find: (a) the transfer function $H(s) = V_o/V_i$, (b) the impulse response, (c) the response when $v_i(t) = u(t)$ V, (d) the response when $v_i(t) = 8 \cos 2t$ V.

Solution:

(a) Using voltage division,

$$V_o = \frac{1}{s+1} V_{ab} \quad (16.9.1)$$

But

$$V_{ab} = \frac{1 \parallel (s+1)}{1 + 1 \parallel (s+1)} V_i = \frac{(s+1)/(s+2)}{1 + (s+1)/(s+2)} V_i$$

or

$$V_{ab} = \frac{s+1}{2s+3} V_i \quad (16.9.2)$$

Substituting Eq. (16.9.2) into Eq. (16.9.1) results in

$$V_o = \frac{V_i}{2s+3}$$

Thus, the transfer function is

$$H(s) = \frac{V_o}{V_i} = \frac{1}{2s+3}$$

(b) We may write $H(s)$ as

$$H(s) = \frac{1}{2} \frac{1}{s + \frac{3}{2}}$$

Its inverse Laplace transform is the required impulse response:

$$h(t) = \frac{1}{2} e^{-3t/2} u(t)$$

(c) When $v_i(t) = u(t)$, $V_i(s) = 1/s$, and

$$V_o(s) = H(s)V_i(s) = \frac{1}{2s(s + \frac{3}{2})} = \frac{A}{s} + \frac{B}{s + \frac{3}{2}}$$

where

$$A = sV_o(s) \Big|_{s=0} = \frac{1}{2(s + \frac{3}{2})} \Big|_{s=0} = \frac{1}{3}$$

$$B = \left(s + \frac{3}{2}\right) V_o(s) \Big|_{s=-3/2} = \frac{1}{2s} \Big|_{s=-3/2} = -\frac{1}{3}$$

Hence, for $v_i(t) = u(t)$,

$$V_o(s) = \frac{1}{3} \left(\frac{1}{s} - \frac{1}{s + \frac{3}{2}} \right)$$

and its inverse Laplace transform is

$$v_o(t) = \frac{1}{3} (1 - e^{-3t/2}) u(t) \text{ V}$$

(d) When $v_i(t) = 8 \cos 2t$, then $V_i(s) = \frac{8s}{s^2 + 4}$, and

$$\begin{aligned} V_o(s) &= H(s)V_i(s) = \frac{4s}{(s + \frac{3}{2})(s^2 + 4)} \\ &= \frac{A}{s + \frac{3}{2}} + \frac{Bs + C}{s^2 + 4} \end{aligned} \quad (16.9.3)$$

where

$$A = \left(s + \frac{3}{2}\right) V_o(s) \Big|_{s=-3/2} = \frac{4s}{s^2 + 4} \Big|_{s=-3/2} = -\frac{24}{25}$$

To get B and C , we multiply Eq. (16.9.3) by $(s + 3/2)(s^2 + 4)$. We get

$$4s = A(s^2 + 4) + B\left(s^2 + \frac{3}{2}s\right) + C\left(s + \frac{3}{2}\right)$$

Equating coefficients,

$$\text{Constant: } 0 = 4A + \frac{3}{2}C \Rightarrow C = -\frac{8}{3}A$$

$$s: \quad 4 = \frac{3}{2}B + C$$

$$s^2: \quad 0 = A + B \Rightarrow B = -A$$

Solving these gives $A = -24/25$, $B = 24/25$, $C = 64/25$. Hence, for $v_i(t) = 8 \cos 2t$ V,

$$V_o(s) = \frac{-24}{25} \frac{1}{s + \frac{3}{2}} + \frac{24}{25} \frac{s}{s^2 + 4} + \frac{32}{25} \frac{2}{s^2 + 4}$$

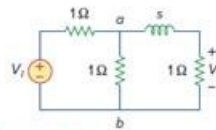


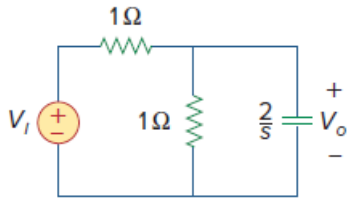
Figure 16.19
For Example 16.9.

$$s + \frac{3}{2} \quad 2s \quad s^2 + 4 \quad 2s \quad s^2 + 4$$

and its inverse is

$$v_o(t) = \frac{24}{25} \left(-e^{-3t/2} + \cos 2t + \frac{4}{3} \sin 2t \right) u(t) \text{ V}$$

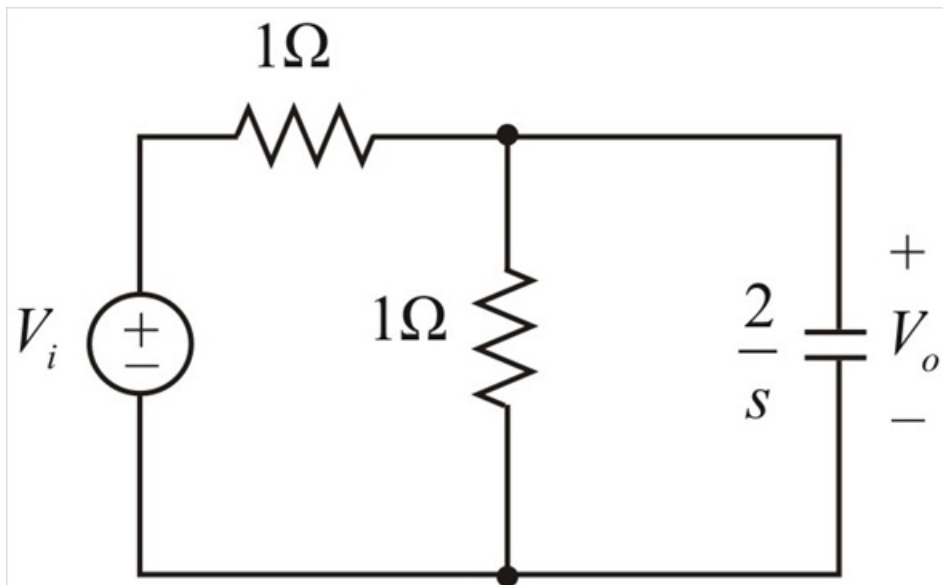
Figure 16.20:



Step-by-step solution

Step 1 of 8

Given circuit



Step 2 of 8

(a) Using Nodal analysis, we get

$$\frac{V_o - V_i}{1} + \frac{V_o}{1} + \frac{V_o}{\frac{2}{s}} = 0$$

$$V_o \left[1 + 1 + \frac{s}{2} \right] = V_i$$

$$V_o \left(2 + \frac{s}{2} \right) = V_i$$

Thus, the transfer function

$$H(s) = \frac{V_o}{V_i}$$

$$\therefore H(s) = \frac{2}{s+4}$$

Step 3 of 8

(b) The impulse response is by taking inverse transform for $H(s)$

$$H(s) = \frac{2}{s+4}$$

$$\therefore \boxed{h(t) = 2e^{-4t}u(t)}$$

Step 4 of 8

(c) When $v_i(t) = u(t)V$

$$V_i(s) = \frac{1}{s}$$

$$V_o(s) = H(s)V_i(s)$$

$$V_o(s) = \left(\frac{2}{s+4}\right)\left(\frac{1}{s}\right)$$

By partial fraction expansion, we get

$$\frac{2}{s(s+4)} = \frac{A}{s} + \frac{B}{s+4}$$

$$2 = A(s+4) + Bs$$

Consider constants

$$4A = 2$$

$$A = \frac{1}{2}$$

Consider $s = -4$

$$(-4)B = 2$$

$$B = \frac{-1}{2}$$

Step 5 of 8

Then,

$$V_o(s) = \frac{1}{2s} - \frac{1}{2(s+4)}$$

Taking inverse transform is

$$v_o(t) = \left(\frac{1}{2} - \frac{1}{2}e^{-4t}\right)u(t) \text{ V}$$

$$\therefore \boxed{v_o(t) = \frac{1}{2}(1 - e^{-4t})u(t) \text{ V}}$$

Step 6 of 8

(d) When $v_i(t) = 8 \cos 2t$

$$V_i(s) = \frac{8s}{s^2 + 4}$$

$$V_o(s) = V_i(s)H(s)$$

$$V_o(s) = \left(\frac{2}{s+4} \right) \left(\frac{8s}{s^2 + 4} \right)$$

By applying partial fraction method

$$\frac{16s}{(s+4)(s^2+4)} = \frac{A}{s+4} + \frac{Bs+C}{s^2+4}$$

$$16s = A(s^2+4) + (Bs+C)(s+4)$$

Consider $s = -4$

$$(16+4)A = (16)(-4)$$

$$A = \frac{-16}{5}$$

Step 7 of 8

$$A = -3.2$$

Consider s^2 – Coefficients

$$A + B = 0$$

$$B = -A$$

$$B = 3.2$$

Consider s – coefficients

$$4B + C = 16$$

$$C = 16 - 4(3.2)$$

$$C = 3.2$$

Step 8 of 8

$$V_o(s) = \frac{-3.2}{s+4} + \frac{3.2s+3.2}{s^2+4}$$

$$V_o(s) = -3.2 \left[\frac{1}{s+4} \right] + 3.2 \left[\frac{s}{s^2+4} \right] + 1.6 \left[\frac{2}{s^2+4} \right]$$

Taking inverse transform, we obtain

$$v_o(t) = (-3.2e^{-4t} + 3.2 \cos(2t) + 1.6 \sin(2t))u(t) \text{ V}$$

$$\therefore \boxed{v_o(t) = 3.2 \left[-e^{-4t} + \cos(2t) + \frac{1}{2} \sin(2t) \right] u(t) \text{ V}}$$